

Seaglider Pilot Guidebook

AA 2026 Capstone Team

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1 Pilot Guidebook Overview

This pilot guidebook serves as the operational reference manual for the Seaglider Damage Assessment Protocol (D.A.P.) system. Its purpose is to provide pilots with a clear, structured procedure for operating the D.A.P. system, interpreting its outputs, and applying the resulting damage assessment information to support informed decision-making.

The D.A.P. system is a diagnostic decision-support tool designed to analyze vehicle response data and identify the most likely damage condition affecting the Seaglider. By comparing measured flight or mission data against predefined fault models, the system evaluates possible failure scenarios and determines the most probable fault case.

This guidebook provides step-by-step operational instructions and interpretation guidance necessary for successful use.

The current DAP system evaluates the following predefined fault scenarios:

- 25% Wing Fault
- 50% Wing Fault
- 50% Rudder Fault

For each diagnostic assessment, the D.A.P. system generates a Final Damage Assessment Report containing the most likely fault diagnosis and supporting comparison metrics. This information enables pilots to quickly assess vehicle condition and determine the most appropriate operational response.

This guidebook includes procedures for launching the D.A.P. system, importing required mission or flight data, executing the diagnostic analysis, interpreting the Final Damage Assessment Report, and understanding system limitations. The document is intended to ensure consistent operation of the D.A.P. system while improving pilot confidence in interpreting diagnostic outputs.

As the D.A.P. system currently operates using a finite set of modeled fault conditions, pilots should recognize that all diagnoses are limited to the predefined fault scenarios available within the current implementation. The output should therefore be used as a decision-support aid in conjunction with pilot judgment and mission awareness.

2 D.A.P. Manual

This section provides step-by-step instructions for operating the D.A.P System in MATLAB. The user shall follow the procedure below to properly execute the D.A.P System using the required mission files.

2.1 Step 1: Access the D.A.P System Folder

Locate and open the Seaglider D.A.P System file folder on your computer. Ensure that all required MATLAB files, scripts, and mission data are contained within the folder before continuing.

Expected Result: The D.A.P System project folder is successfully opened and accessible.

2.2 Step 2: Open the DAP_Main File in MATLAB

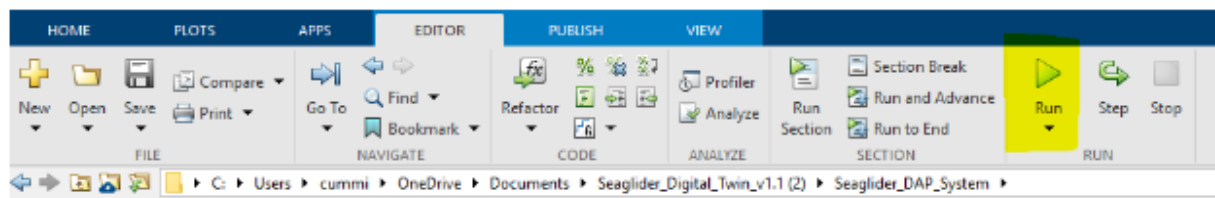
Open MATLAB and navigate to the D.A.P System folder. Locate the file named DAP_Main and open it in the MATLAB Editor.

Name	Status	Date modified	Type	Size
MS_Comparator_Plots	✓	5/6/2026 12:22 PM	File folder	
coefficientsV3	✓	5/5/2026 10:16 AM	MATLAB Code	2 KB
coefficientsV4_nominal	✓	5/14/2026 3:54 PM	MATLAB Code	14 KB
Create_Mission_Data_Matrix	✓	5/14/2026 2:51 PM	MATLAB Code	6 KB
DAP_Main.asv	✓	5/8/2026 2:34 PM	ASV File	5 KB
DAP_Main	✓	5/14/2026 3:55 PM	MATLAB Code	5 KB
Digital_Twin.asv	✓	5/11/2026 11:09 AM	ASV File	5 KB
Digital_Twin	✓	5/14/2026 2:56 PM	MATLAB Code	8 KB
dynamicsV4	✓	5/14/2026 2:53 PM	MATLAB Code	9 KB
Log_File_Unpacker	✓	5/6/2026 3:45 PM	MATLAB Code	4 KB
NC_File_Unpacker	✓	5/6/2026 4:55 PM	MATLAB Code	3 KB
p1750005	✓	5/6/2026 5:05 PM	Text Document	22 KB
p1750005	✓	5/6/2026 5:05 PM	NC File	661 KB
p1950004	✓	5/5/2026 10:16 AM	Text Document	10 KB
p1950004	✓	5/6/2026 12:42 PM	NC File	822 KB
parametersV3	✓	5/5/2026 10:16 AM	MATLAB Code	2 KB

Expected Result: The DAP_Main script is opened in the MATLAB Editor window.

2.3 Step 3: Run the D.A.P System Script

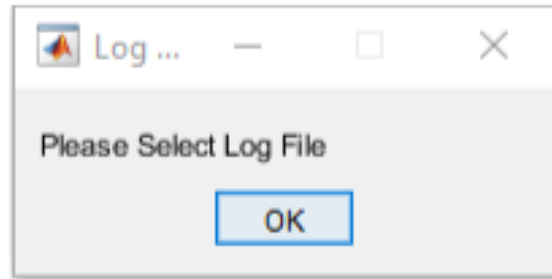
In the MATLAB Editor tab, press the green **Run** button to begin execution of the D.A.P System code.



Expected Result: MATLAB begins execution of the D.A.P System script.

2.4 Step 4: Select the Required Log File

After pressing the Run button, a popup window will appear prompting the user to:



Press **OK** to open the file explorer window.

Navigate to the location of the desired .log mission data file.

Select the desired .log file by either:

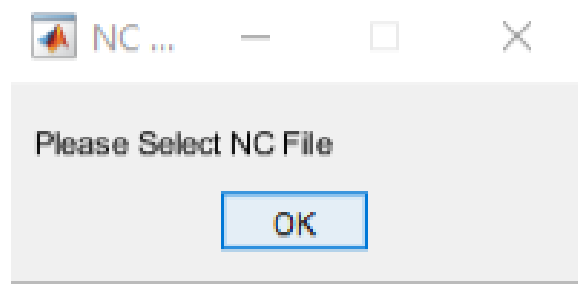
- Double-clicking the file, or
- Single-clicking the file and pressing the **Open** button in the lower left corner of the file explorer window.

Name	Status	Date modified	Type	Size
MS_Comparator_Plots	✓	5/6/2026 12:22 PM	File folder	
p1750005	✓	5/6/2026 5:05 PM	Text Document	22 KB
p1950004	✓	5/5/2026 10:16 AM	Text Document	10 KB

Expected Result: The selected .log file is successfully loaded into the D.A.P System.

2.5 Step 5: Select the Required NC File

After selecting the .log file, a second popup window will appear prompting the user to:



Press **OK** to open the file explorer window.

Navigate to the location of the desired .nc mission data file.

Select the desired .nc file by either:

- Double-clicking the file, or
- Single-clicking the file and pressing the **Open** button in the lower left corner of the file explorer window.

Name	Status	Date modified	Type	Size
MS_Comperator_Plots	🟢	5/6/2026 12:22 PM	File folder	
p1750003	🟢	5/6/2026 5:05 PM	NC File	661 KB
p1950004	🟢	5/6/2026 12:42 PM	NC File	822 KB

Expected Result: The selected .nc file is successfully loaded into the D.A.P System.

2.6 Step 6: D.A.P System Execution

Once both the .log and .nc files have been selected, MATLAB will automatically continue execution of the D.A.P System code.

The D.A.P System will process the provided mission data and generate the corresponding fault analysis outputs.

=====	
FINAL DAMAGE ASSESSMENT REPORT	
=====	
Primary Diagnosis: Nominal Case	
Difference Metric: 60.35%	
Fault_Case	Data_Above_Margin
-----	-----
{'Nominal Case' }	60.347
{'50% Rudder Fault'}	76.281
{'25% Wing Fault' }	87.014
{'50% Wing Fault' }	87.633

Expected Result: The D.A.P System successfully executes and generates percentage-based fault analysis results.

3 D.A.P. Output and Interpretation

After executing the D.A.P. script in MATLAB, the system generates a *Final Damage Assessment Report*, which provides the pilot with the system's assessment of the most likely damage condition based on the uploaded flight or test data. This report is the final output of the Damage Assessment Protocol (DAP) and is intended to help the pilot quickly identify the fault condition affecting the vehicle.

The current DAP system evaluates three predefined fault cases:

- 25% Wing Fault
- 50% Wing Fault
- 50% Rudder Fault

For each of these fault cases, the system compares the measured vehicle response against the expected response generated by the corresponding fault model. The output report contains two primary components that the pilot must understand:

- **Primary Diagnosis**
- **Percent Error for Each Fault Case**

3.1 Primary Diagnosis

The **Primary Diagnosis** is the fault condition identified by the D.A.P. system as the most likely explanation for the observed vehicle behavior. This diagnosis is determined by comparing the average percent error values for all available fault cases and selecting the fault model with the smallest error.

This represents the D.A.P. system's final recommendation for the most probable damage condition affecting the Seaglider.

3.2 Percent Error

The **Percent Error** quantifies how closely the actual measured system behavior matches the predicted behavior for each fault model.

A lower percent error indicates that the measured vehicle response is a closer match to that fault case. Conversely, a higher percent error indicates a poorer match.

The interpretation is straightforward:

- **Error close to 0%** = Strong match to that fault condition
- **Moderate error** = Partial similarity to that fault condition
- **High error** = Weak match to that fault condition

Because the D.A.P. system compares all available fault cases simultaneously, the fault case with the lowest percent error is identified as the most likely diagnosis.

3.3 Example Interpretation

Consider the following example output:

- 50% Wing Fault: 0.0%
- 25% Wing Fault: 62.4%
- 50% Rudder Fault: 74.8%

In this case, the *50% Wing Fault* produces an average percent error of 0.0%, meaning the measured vehicle response perfectly matches the predicted response for that fault model.

The other two fault cases produce significantly larger errors, indicating that their predicted responses do not align closely with the observed data.

Since the 50% Wing Fault has the lowest error value by a substantial margin, the D.A.P. system identifies:

Primary Diagnosis: 50% Wing Fault

This would indicate a high-confidence diagnosis because there is a clear separation between the most likely fault case and the alternatives.

4 Understanding D.A.P. Results

This section explains how the Damage Assessment Protocol (D.A.P.) processes mission data, detects abnormal vehicle behavior, isolates potential faults, and generates the final damage assessment report.

The goal of this section is to help pilots and future operators understand not only what the D.A.P. output means, but also how the diagnosis was produced.

4.1 Overview of the Diagnostic Process

The D.A.P. system follows the workflow shown in Figure X.

1. Import mission data from the selected Log and NC files.
2. Create a Mission Data Matrix containing the vehicle data used by the system.
3. Generate a nominal Digital Twin simulation.
4. Compare the mission data against the nominal Digital Twin.
5. Identify variables exhibiting sustained abnormal behavior.
6. Generate off-nominal fault simulations.
7. Compare each fault simulation against the mission data.
8. Calculate an Average Percent Error for each fault case.
9. Select the fault case with the lowest Average Percent Error as the Primary Diagnosis.

4.2 Mission Data Matrix

The first step in the D.A.P. process is the creation of a Mission Data Matrix.

The Mission Data Matrix combines selected information from both the mission Log file and NC file into a single dataset that is used throughout the diagnostic process.

The current D.A.P. implementation utilizes the following mission variables:

- eng_depth
- eng_head
- eng_pitchAng
- eng_rollAng
- eng_pitchCtl
- eng_rollCtl
- eng_vbdCC
- vert_speed_gsm
- temperature
- pressure
- density

Several important vehicle parameters stored in the Log file are also included within the Mission Data Matrix.

If missing values are encountered, the D.A.P. system automatically fills those values to create a continuous dataset suitable for comparison and simulation.

Table 1: Example Mission Data Matrix Structure

Time	Depth	Heading	Pitch Angle	Roll Angle	Pitch Control	Roll Control	Vertical Speed
eng_elaps_t	eng_depth	eng_head	eng_pitchAng	eng_rollAng	eng_pitchCtl	eng_rollCtl	vert_speed_gsm
0.0	0.0	182.4	-12.1	1.8	-0.45	0.02	-6.3
10.0	4.7	183.1	-13.5	2.1	-0.48	0.03	-6.8
20.0	9.6	184.0	-14.0	2.4	-0.50	0.05	-7.1
30.0	14.2	185.6	-13.8	2.8	-0.49	0.06	-6.9

The Mission Data Matrix is the dataset used by the D.A.P. system during comparison. Each column represents a mission variable, and each row represents one recorded time step from the Seaglider mission.

The values shown in Table 1 are example values only. In the actual D.A.P. system, this matrix is automatically created from the selected Log and NC files.

4.3 Comparing Mission Data to the Digital Twin

Once the Mission Data Matrix is created, the D.A.P. system generates a nominal Digital Twin simulation.

The Digital Twin is a simulated version of a healthy Seaglider operating under the same mission conditions. The D.A.P. system uses this simulation as a baseline for comparison against the actual mission data.

Before the comparison begins, the D.A.P. system removes:

- NaN values
- Infinite values
- Zero-value comparison points when required for percent-difference calculations

The D.A.P. system then focuses only on the middle 60% of the mission.

The first 20% and final 20% of the mission are intentionally excluded from analysis. This reduces the likelihood of false detections caused by launch conditions, mission initialization, recovery procedures, or end-of-mission transients.

4.4 Acceptable Differences Between Mission Data and the Digital Twin

Not every difference between the mission data and the Digital Twin indicates a fault.

Each variable has an allowable tolerance that defines how much deviation is considered acceptable before the behavior is classified as abnormal.

The current tolerance values are:

Table 2: Current D.A.P. Variable Tolerances

Variable	Tolerance (%)
eng_pitchAng	7.5
eng_rollAng	187.5
eng_head	13.5
eng_depth	14.5
All Other Variables	2.0

These tolerances define the acceptable difference between the mission data and the Digital Twin simulation.

4.5 Percent Difference Calculation

For each variable, the D.A.P. system calculates the percent difference between the mission data and the Digital Twin simulation using:

$$E = \left| \frac{MI - DT}{DT} \right| \times 100 \quad (1)$$

where:

- MI = Mission Data
- DT = Digital Twin Data
- E = Percent Difference

If the percent difference exceeds the allowable tolerance for that variable, the data point is classified as an error.

If the percent difference remains below the tolerance, the data point is considered acceptable.

4.6 Window-Based Failure Detection

The D.A.P. system does not determine failures from individual data points.

Instead, the mission data is divided into windows containing twenty samples each. The D.A.P. system then determines how many samples within each window exceed the allowable tolerance.

For each window:

$$Window\ Error\ Rate = \frac{\text{Number of Samples Outside Tolerance}}{\text{Total Samples in Window}} \times 100 \quad (2)$$

A window is flagged when:

More than 40% of the samples within that window exceed the allowable tolerance.

The current threshold is defined in:

```
missioncompare_version_4.m
```

as:

```
failureThreshold = 40;
```

The intended behavior is that nominal mission cases remain below the 40% threshold, while fault cases exceed the threshold in a sustained manner.

This value can be modified in future versions if testing indicates that the detector is either too sensitive or not sensitive enough.

4.7 Sustained Failure

A variable is not classified as failed simply because one analysis window exceeds the 40% threshold.

Instead, the D.A.P. system applies a sustained-failure rule.

A variable only fails when:

- A window exceeds the 40% threshold, and
- Every remaining window after that point also exceeds the threshold

or

- The final analysis window exceeds the threshold.

This logic prevents isolated spikes, temporary disturbances, or short-duration anomalies from being incorrectly identified as faults.

The D.A.P. system is designed to detect sustained abnormal behavior rather than isolated events.

4.8 Understanding Diff and Window Scores

When a failure is detected, the D.A.P. output displays a Diff score.

Example:

```
eng_pitchAng failed with Diff (max window score) = 100.00
```

The Diff value represents the largest window error rate observed during the comparison process.

A Diff score of:

- 0 indicates that no window exceeded the allowable tolerance.
- 40 indicates that the worst window contained approximately 40% error samples.
- 100 indicates that every sample within at least one analysis window exceeded the allowable tolerance.

A Diff score of 100 does not mean the pitch angle was 100% incorrect.

Instead, it means that every sample within the worst analysis window exceeded the allowable tolerance for that variable.

4.9 Determining the Most Likely Fault

If sustained abnormal behavior is detected, the D.A.P. system proceeds to the fault-isolation stage.

The current implementation evaluates three fault cases:

- 25% Wing Fault
- 50% Wing Fault
- 50% Rudder Fault

A Digital Twin simulation is generated for each fault case.

Each fault simulation is then compared against the mission data using the same comparison process described previously.

For every variable, the D.A.P. system calculates the mean percent difference between the mission data and the fault simulation.

These values are then averaged together to produce a single metric called the Percent Error.

4.10 Understanding Percent Error

The Percent Error represents how closely a fault-case simulation matches the actual mission data.

Lower values indicate stronger agreement between:

- The recorded mission data
- The fault-case simulation

Higher values indicate poorer agreement.

The D.A.P. system ranks all fault cases from lowest Percent Error to highest Percent Error.

4.11 Understanding the Primary Diagnosis

The fault case with the lowest Percent Error is selected as the Primary Diagnosis.

Example:

Primary Diagnosis: 25% Wing Fault Percent Error: 18.42%

This output indicates that the 25% Wing Fault simulation most closely matched the recorded mission data and was therefore identified as the most likely fault condition.

4.12 Using seaglider.pub to Verify D.A.P. Results

The D.A.P. output identifies where abnormal behavior was detected. Pilots should then use seaglider.pub to examine the corresponding mission plots and determine whether the observed vehicle behavior supports the reported diagnosis.

The variables reported by D.A.P. correspond directly to data visualizations available on seaglider.pub.

For example:

- **eng_pitchAng**

Review the Pitch Angle plot. Look for excessive oscillations, unstable dive-to-climb transitions, or persistent pitch offsets.

- **eng_rollAng**

Review the Roll Angle plot. Look for asymmetric rolling behavior, excessive roll angles, or sustained rolling tendencies.

- **eng_head**

Review the Heading plot and mission track. Look for course deviations, steering instability, or difficulty maintaining the intended heading.

- **eng_depth**

Review the Depth profile. Look for abnormal dive trajectories, inconsistent climb behavior, or unexpected depth changes.

- **vert_speed_gsm**

Review the Vertical Velocity plot. Look for reduced climb rates, reduced descent rates, or unstable vertical motion.

The D.A.P. output identifies which variables contributed to the diagnosis, while seaglider.pub provides the visual evidence necessary to verify and better understand the detected behavior.

Table 3: D.A.P. Output Interpretation and Verification Matrix

D.A.P. Variable	seaglider.pub Plot	What D.A.P. Detected	What the Pilot Should Look For
eng_pitchAng	Pitch Angle	Pitch behavior differs from the Digital Twin prediction	Excessive oscillations, unstable dive-to-climb transitions, persistent nose-up or nose-down offsets
eng_rollAng	Roll Angle	Roll behavior differs from the Digital Twin prediction	Sustained rolling to one side, excessive roll angles, asymmetric behavior during dives or climbs
eng_head	Heading / Mission Track	Heading behavior differs from the Digital Twin prediction	Course deviations, steering instability, excessive heading corrections, inability to maintain intended heading
eng_depth	Depth Profile	Depth behavior differs from the Digital Twin prediction	Unexpected depth changes, abnormal dive profiles, inconsistent climb or descent behavior
vert_speed_gsm	Vertical Velocity	Vertical speed differs from the Digital Twin prediction	Reduced climb rate, reduced descent rate, unstable vertical motion, degraded glide performance